

Charleston Southern University | CSCI 496 Senior Project Defense

GAMIFIED HABIT TRACKER

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STATEMENT OF PURPOSE

Problem Statement

- Users struggle to maintain long-term habits due to inconsistent motivation. Traditional habit-tracking apps lack effective gamification, leading to disengagement and abandonment

Project Purpose

- Build a habit tracker that uses gamification element, such as experience points, streaks, and achievements, to make habit formation engaging and sustainable

Project Goal

- Develop a functional application using Flutter and Firebase that stores user data, tracks progress, and rewards consistent behavior through a clean, and simple UI.

RESEARCH AND BACKGROUND

Habit Formation

Research shows that consistent repetition and tracking are essential for building long-term habits, but users often struggle to maintain motivation over time

Gamification

Studies suggest that elements like points, rewards, streaks, and achievements can significantly increase user engagement and encourage consistent behavior

Existing Solutions

Current habit-tracking apps provide basic tracking features, but many lack strong gamification systems, leading to decreased long-term user retention

Technology

Modern tools like Flutter and Firebase enable scalable, cross-platform mobile applications with real time data syncing and secure user authentication

PROJECT LANGUAGES, SOFTWARE, AND HARDWARE

SOFTWARE

- Flutter SDK
- FirebaseAuthentication
- Cloud Firestore
- Github

HARDWARE

- Personal laptop
- iPhone
- Internet connection
- External storage

LANGUAGE

- Dart (Flutter)

PROJECT REQUIREMENTS

Functional

- User can create, edit, and delete habits
- Users can mark habits as complete each day
- System tracks experience points and updates them
- System tracks habit streaks
- Users can unlock and view achievements based on progress
- Users can view habit progress and analytics

Non-Functional

- System should load the main dashboard withing 3-5 seconds
- System should update XP and streaks instantly after completion
- System should securely store user data using Firebase authentication
- Application should be easy to use with simple navigations and UI
- Application should be reliable and not crash when closed or reopened

PROJECT IMPLEMENTATION DESCRIPTION AND DEMONSTRATION

Habit Love is a gamified habit tracking application built with Flutter and Firebase that allows users to create and complete daily habits while earning XP, leveling up, and unlocking achievements, featuring five main screens including a Dashboard (Fig. 1), Habit List (Fig. 2), Analytics (Fig. 3), Achievements (Fig. 4), and Profile (Fig. 5). The application is built using modular Flutter architecture with Firebase serving as the backend for authentication and real-time data storage.

A live demo will be presented to show the main features of the app in action

https://github.com/mskegro/habit_tracker

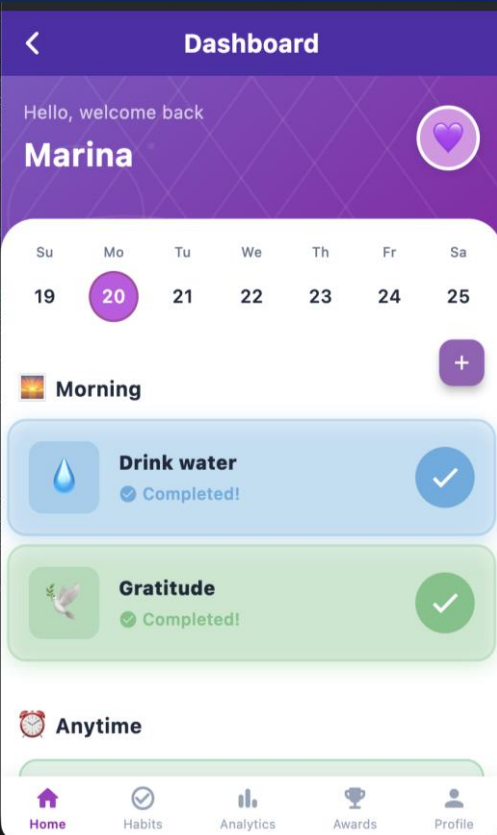


Fig 1. Dashboard Screen

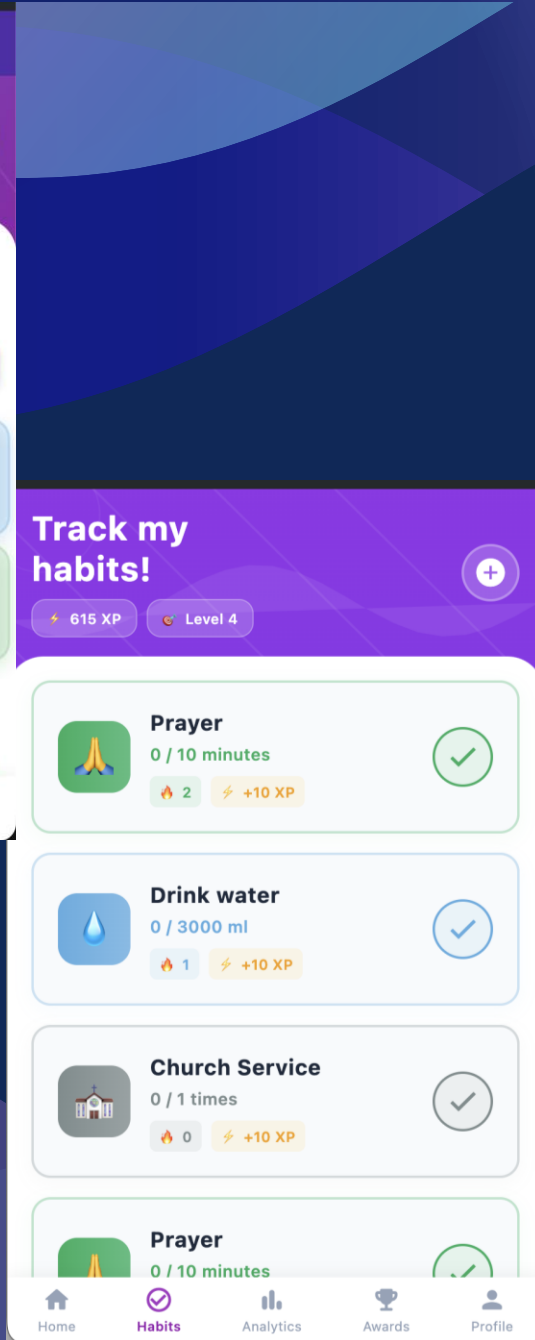


Fig 2. Habit Detail Screen

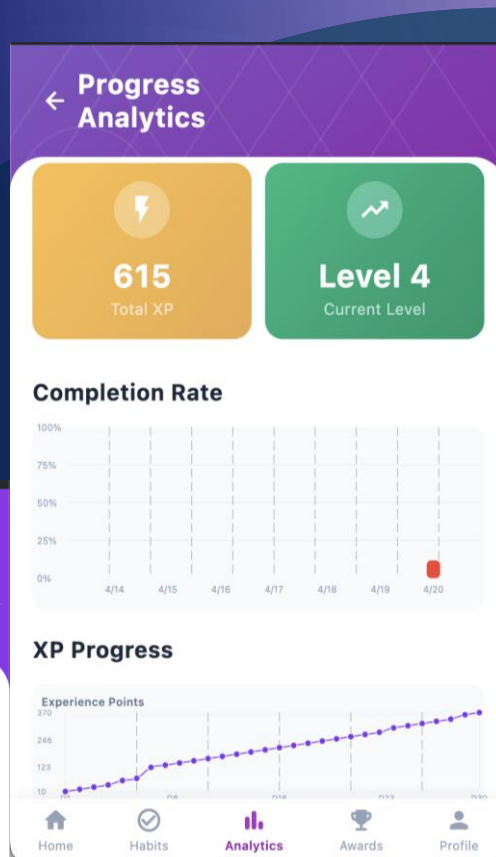


Fig 3. Analytics Screen

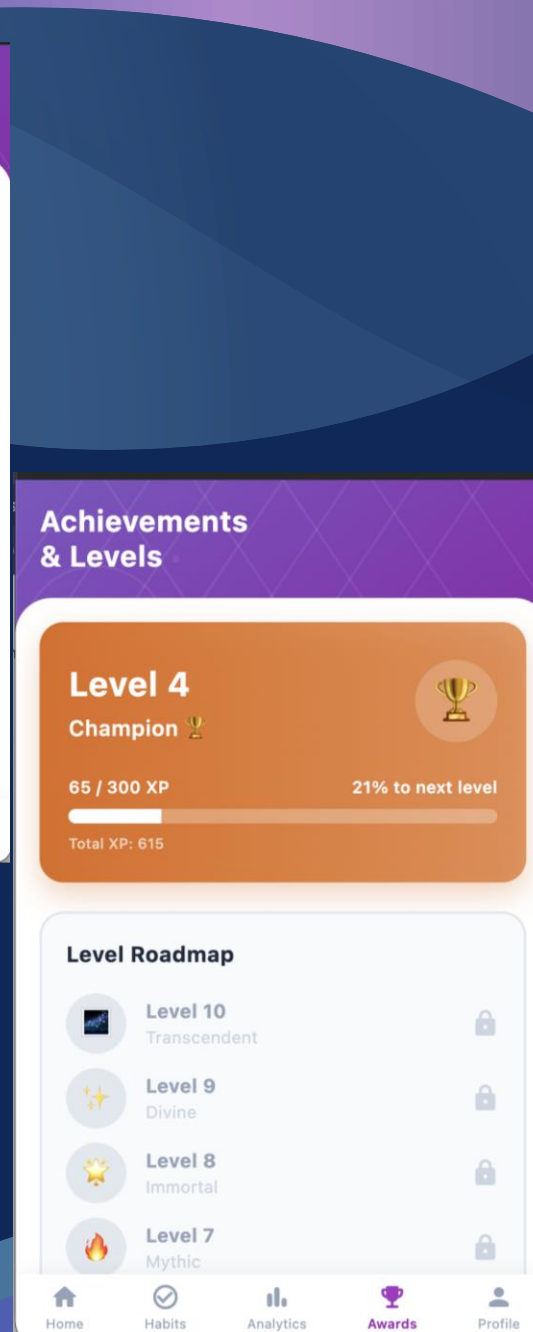


Fig 4. Achievements screen

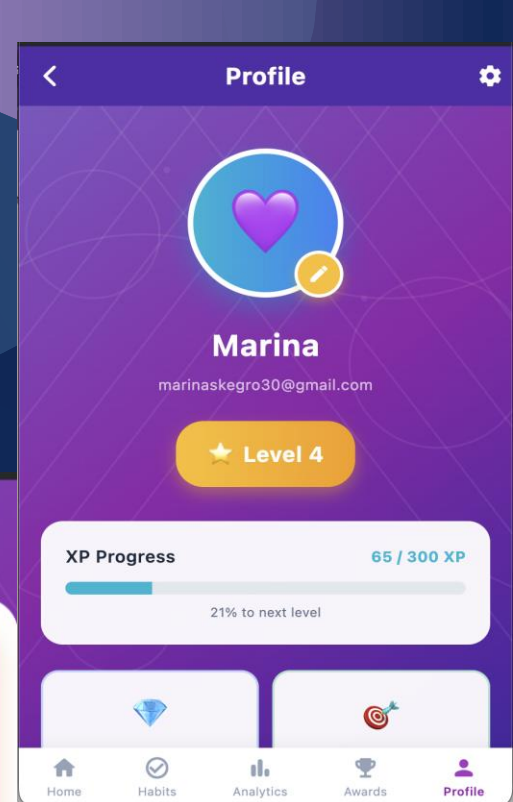


Fig 5. Profile Screen

TEST PLAN

Functional Testing

- Tested habit creation, editing, and deletion
- Verified XP system updates correctly after actions
- Checked streak tracking and achievement unlocking

Security Testing

- Verified Firebase authentication for secure login
- Ensured users can only access their own data
- Data storage is securely handled in Firebase

Performance Testing

- Measured app load time
- Tested performance with increasing number of habits
- Verified smooth navigation and fast response times

Usability Testing

- Tested simple and intuitive navigation
- Dashboard gives quick access to main features
- Checked clear and readable UI design

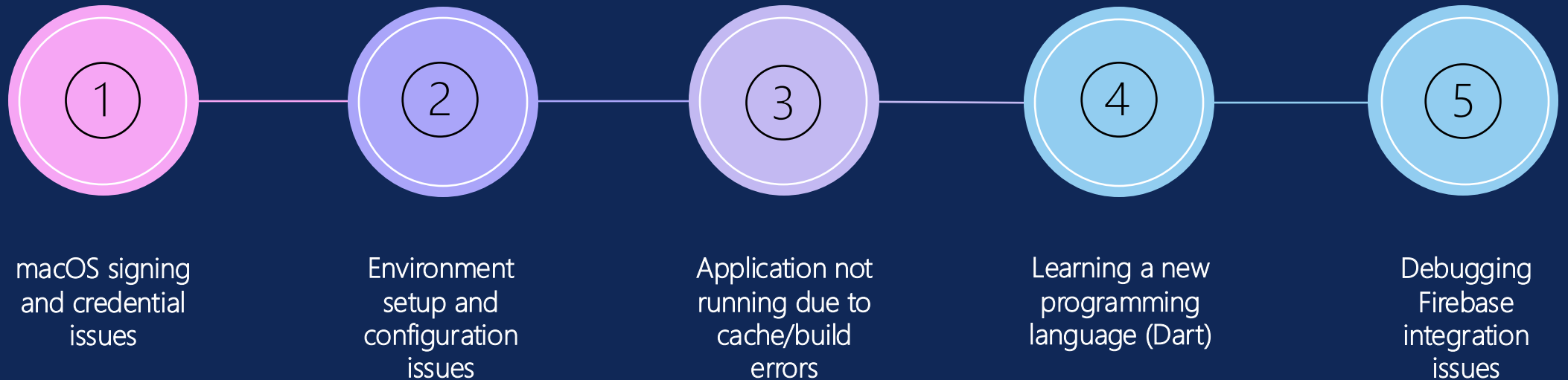
TEST RESULTS

Test Type	Test Case Summary	Result	Notes
Functional	Tested core features such as habit management, XP system, streak tracking, achievements, analytics, and Firebase sync	Pass	All functional requirements worked correctly
Performance	Tested app load time, responsiveness, scalability, update speed, and stability under heavy usage	Pass	App performed smoothly with no lag or delays
Security	Tested Firebase authentication, data access control, login methods, and secure data storage	Pass	All security rules and authentication systems worked correctly
Usability	Tested navigation, UI clarity, accessibility	Pass	Interface was intuitive, clear, and user-friendly

USER TESTING RESULTS

- 8 participants
- 7-day testing period
- SUS Score: 76/100
- Feature Satisfaction: 4.3/5.0
- 7/8 participants would continue using the app
- What worked:
 - Users liked the app's simplicity and ease of use
 - Habit creation and XP system were intuitive and enjoyable
 - Overall navigation and UI received positive feedback
- Area for Improvement
 - Notifications and reminders would improve daily accountability.

CHALLENGES OVERCOME



FUTURE ENHANCEMENTS

Smart Notifications

- Add personalized reminders based on user habits and activity patterns

Daily Challenges

- Introduce challenges to increase motivation and user engagement

Social Features

- Allow friends, groups, or leaderboards for competition and accountability

Advanced Analytics

- Provide deeper insights like trends, progress predictions, and performance tracking

Customization Options

- Let users personalize themes, goals, and reward systems

Offline Mode Support

- Allow users to track habits and view progress without an internet connection, syncing data once back online



THANK YOU

Marina Skegro

https://github.com/mskegro/habit_tracker